

GRENCHE, Switzerland

WMO: 066320

Lat: 47.1792N

Lon: 7.4150E

Elev: 1407

StdP: 13.96

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.5	20.9	8.2	8.8	18.2	16.3	13.2	22.9	26.4	37.1	23.5	35.0	9.8	60	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	88.1	68.7	84.8	67.9	81.4	66.3	70.2	84.0	68.8	81.7	67.4	78.8	7.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	101.3	74.0	64.7	96.5	72.1	63.5	92.5	71.0	35.0	84.2	33.7	82.2	32.6	79.4	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.4	18.3	15.4	DB	10.5	93.3	5.4	3.1	6.6	95.5	3.4	97.3	0.4	99.1	-3.5	101.4	(4)
(5)			WB	10.0	73.9	5.5	2.2	6.0	75.4	2.8	76.7	-0.3	77.9	-4.4	79.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	34.5	34.7	42.9	50.3	56.3	64.6	67.8	67.2	59.9	51.4	41.9	35.2	(6)	
	DBStd	13.54	6.38	7.30	5.72	6.29	6.20	5.77	5.44	5.58	5.22	5.89	5.58	6.31	(7)	
	HDD50	1988	482	428	228	73	14	0	0	0	0	56	248	459	(8)	
	HDD65	5559	946	847	687	443	275	78	33	43	169	421	693	924	(9)	
	CDD50	2224	0	1	6	81	211	438	553	532	297	100	5	0	(10)	
	CDD65	318	0	0	0	0	6	65	121	110	16	0	0	0	(11)	
	CDH74	3256	0	0	0	16	117	716	1214	1033	160	0	0	0	(12)	
	CDH80	998	0	0	0	1	12	219	413	337	16	0	0	0	(13)	
(14) Wind	WSAvg	5.2	6.0	6.0	6.2	5.5	5.5	5.1	4.9	4.0	4.5	4.2	4.6	5.3	(14)	
(15) Precipitation	PrecAvg	47.4	3.4	3.3	3.1	3.6	4.3	4.4	4.7	4.5	3.9	3.9	3.9	4.2	(15)	
	PrecMax	60.2	9.6	8.2	11.9	7.6	8.0	9.6	9.8	10.4	7.0	7.4	9.1	9.3	(16)	
	PrecMin	33.3	0.6	0.9	0.9	0.5	1.4	2.0	1.8	0.4	1.7	0.6	0.4	0.0	(17)	
	PrecStd	7.8	2.2	1.9	2.5	2.0	1.9	1.9	1.7	2.1	1.7	1.8	2.3	2.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.8	58.8	66.9	76.5	82.0	90.3	94.0	90.7	82.1	71.8	61.1	51.8	(19)	
		MCWB	48.3	48.1	51.6	60.2	64.2	72.2	69.6	69.9	65.4	59.1	54.1	48.5	(20)	
	2%	DB	48.6	52.3	63.7	71.8	77.2	86.1	88.4	87.3	79.1	68.0	56.6	49.4	(21)	
		MCWB	46.1	45.3	50.5	56.6	61.6	69.3	68.0	69.1	64.9	58.5	52.4	47.3	(22)	
	5%	DB	45.5	48.2	59.5	68.3	73.7	81.7	84.7	84.1	74.9	64.1	53.4	47.1	(23)	
		MCWB	43.1	43.6	48.8	55.5	60.0	67.0	67.3	67.8	62.9	57.4	50.9	45.1	(24)	
	10%	DB	43.1	44.5	54.9	64.1	69.8	77.9	80.8	80.1	71.1	60.9	49.9	44.3	(25)	
		MCWB	41.2	41.0	46.9	53.1	58.4	64.9	65.7	66.7	61.6	56.3	48.0	42.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.3	49.7	53.0	59.8	65.8	74.0	71.3	71.6	67.6	61.9	55.3	49.4	(27)	
		MCDB	50.7	54.9	64.0	73.5	78.1	87.9	87.4	85.6	77.1	67.4	59.0	50.5	(28)	
	2%	WB	46.5	46.2	51.0	57.6	63.2	70.5	69.5	70.1	66.0	60.0	53.1	47.6	(29)	
		MCDB	48.4	52.1	60.9	69.7	74.1	82.7	84.4	83.7	76.1	64.8	55.9	49.1	(30)	
	5%	WB	43.4	43.7	49.4	55.8	61.1	68.2	68.1	68.8	64.3	58.4	50.8	45.3	(31)	
		MCDB	45.1	47.8	58.1	67.0	70.8	79.5	81.5	81.4	72.5	62.5	53.1	46.9	(32)	
	10%	WB	41.5	41.2	47.5	53.8	59.3	66.0	66.8	67.3	62.5	56.8	48.2	42.8	(33)	
		MCDB	43.0	44.1	54.2	62.6	68.0	75.8	78.1	78.1	69.4	60.4	49.7	44.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	12.9	19.0	20.4	19.7	20.2	21.1	21.1	19.4	15.4	10.1	8.9	(35)	
		MCDBR	12.9	21.7	29.3	29.3	27.5	28.0	28.8	27.1	25.5	21.8	16.3	12.3	(36)	
	5% WB	MCWBR	10.1	15.0	16.9	15.3	13.2	12.4	10.3	10.2	12.4	13.1	11.9	10.3	(37)	
		MCWBR	12.5	20.3	25.9	27.2	24.8	26.2	25.5	25.3	21.9	18.7	14.4	11.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.300	0.326	0.362	0.400	0.410	0.413	0.413	0.404	0.382	0.358	0.323	0.297	(40)	
		taud	2.440	2.366	2.312	2.231	2.249	2.286	2.280	2.316	2.360	2.409	2.455	2.447	(41)	
	Ebn at Noon		251	266	271	270	271	271	269	266	261	250	239	237	(42)	
		Edh at Noon	21	28	35	41	42	41	40	37	33	27	21	19	(43)	
(44) All-Sky Solar Radiation	RadAvg		371	636	1017	1380	1595	1888	1820	1582	1212	768	415	301	(44)	
	RadStd		47	99	154	204	152	175	161	148	113	74	52	55	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (111 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+1.35	+0.72	+0.54	N/A	N/A	+103	+31

Nomenclature: See separate page